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Shaft sealing structure

Abstract

A shaft sealing structure includes a V-packing; a first guide bush attached to an inner circumference of the holder; and a scraper arranged at an end of the holder. The V-packing is formed of a thick-walled top portion, a thin-walled inner side skirt portion extending diagonally inward from the top portion, and a thin-walled outer side skirt portion extending diagonally outward from the top portion. The inner side skirt portion slidably contacts an outer surface of the rod at a distal end of the inner side skirt portion, and the outer side skirt portion abuts against an inner surface of the holder at a distal end of the outer side skirt portion. The top portion is sandwiched between the scraper and the first guide bush, and the inner and outer side skirt portions are freely deformable inside a space formed between the first guide bush and the scraper.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of and claims benefit under 35 U.S.C. § 120 to U.S. application Ser. No. 17/655,592, filed Mar. 21, 2022, and claims the benefit of priority under 35 U.S.C. § 119 from Japanese Patent Application No. 2022-002691, filed Jan. 12, 2022, and claims the benefit of priority under 35 U.S.C. § 119 from Japanese Patent Application No. 2021-048030, filed, Mar. 23, 2021, the entire contents of each of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

- (1) The present invention relates to a shaft sealing structure arranged on a sliding portion of a rod.
- Description of the Related Art
- (2) Conventionally, a technology has been known in which a shaft seal formed of V-packings are arranged between a rod operating within a liquid such as a chemical or steam, and a body in which the rod is accommodated. For example, in JP H04-049282 U, as a valve rod seal for providing sealing between a valve rod and a valve body, a structure is disclosed in which a V-packing made of a fluorine rubber is sandwiched between V-packings made of a fluorine resin. Consequently, attempts are being undertaken to secure a wider range of sealing performance from low temperatures to high temperatures. Such V-packings are pressed strongly against the valve rod and the valve body due to a reaction force of the spring.
- (3) Further, for example, in JP H03-096468 U, as packings that are mounted between a piston and a cylinder head of a fluid pressure feeding device, a structure is disclosed in which V-packings made of a fluorine resin are stacked in multiple stages. In each of the V-packings, the thickness of the top portion thereof is thinner than that of the other portions, and a rear side apex angle thereof is smaller than a front side apex angle. When the V-packings are stacked in multiple stages and pressed together, an amount by which the V-packings protrude to an outer side can be increased, and an effect of preventing liquid leakage can be stably obtained.

SUMMARY OF THE INVENTION

(4) However, the V-packings of this type which are stacked in multiple stages are formed with a structure in which the sealing portions of the packings are strongly pressed against a target member while being mutually pressed in contact with each other, which results in a large amount of wear on the sealing portions. Further, wearing of all of the V-packings progresses simultaneously. Therefore, there is a concern that the useful lifetime of the shaft seal may be shortened.

(5) The present invention has the object of solving the aforementioned problem.

(6) A shaft sealing structure according to the present invention is arranged between a rod configured to be driven within a predetermined fluid and move in an axial direction, and a holder in which the rod is accommodated. The shaft sealing structure comprises a V-packing; a first guide bush attached to an inner circumference of the holder; and a scraper arranged at an end of the holder. The V-packing is formed of a top portion formed to be thick-walled, an inner side skirt portion formed to be thin-walled and extending diagonally inward from the top portion, and an outer side skirt portion formed to be thin-walled and extending diagonally outward from the top portion. The inner side skirt portion slidably contacts an outer surface of the rod at a distal end of the inner side skirt portion, and the outer side skirt portion abuts against an inner surface of the holder at a distal end of the outer side skirt portion. The top portion is sandwiched between the scraper and the first guide bush, and the inner side skirt portion and the outer side skirt portion are freely deformable inside a space formed between the first guide bush and the scraper.

(7) Further, the top portion includes a convex portion configured to protrude to a side opposite to the inner side skirt portion and the outer side skirt portion, the first guide bush is equipped with a recessed portion in which the upper V-packing is accommodated, and abuts against the convex portion of the V-packing on a bottom surface of the recessed portion, and an end of the scraper abuts against a rear surface of the top portion of the V-packing.

(8) In accordance with the above-described shaft sealing structures, by the inner side skirt portions and the outer side skirt portions receiving a fluid pressure and being elastically deformed thereby, the V-packings or the U-packings can exhibit satisfactory sealing performance. Further, since wearing of the plurality of V-packings or the plurality of U-packings does not progress simultaneously, satisfactory sealing performance can be maintained over a prolonged period of time.

(9) The shaft sealing structures according to the present invention include the plurality of V-packings or the plurality of U-packings that are formed of the thick-walled top portion, the thin-walled inner side skirt portion, and the thin-walled outer side skirt portion, wherein the adjacent ones of the inner side skirt portions do not interfere with each other, and the adjacent ones of the outer side skirt portions do not interfere with each other. Therefore, satisfactory sealing performance can be maintained over a prolonged period of time.

(10) The above and other objects, features, and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings, in which preferred embodiments of the present invention are shown by way of illustrative example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an external view of an angle seat valve in which a shaft sealing structure according to a first embodiment of the present invention is adopted;

(2) FIG. 2 is a cross-sectional view taken along line II-II of the angle seat valve shown in FIG. 1;

(3) FIG. 3 is an enlarged view of the shaft sealing structure in the angle seat valve shown in FIG. 2;

(4) FIG. 4 is a cross-sectional view of a V-packing in the shaft sealing structure shown in FIG. 3;

- (5) FIG. 5 is a view corresponding to FIG. 2 at a time when the angle seat valve shown in FIG. 1 is in a predetermined operational state;
- (6) FIG. 6 is a view corresponding to FIG. 3 and showing a shaft sealing structure according to a second embodiment of the present invention;
- (7) FIG. 7 is a cross-sectional view of a U-packing in the shaft sealing structure shown in FIG. 6; and
- (8) FIG. 8 is a cross-sectional view of a shaft sealing structure according to a reference example.

DESCRIPTION OF THE INVENTION

(9) In the following description, when terms related to upper and lower directions are used, such terms refer to directions on the drawing, and the upper directions including the meaning of “diagonally up” are simply referred to as “up (upper),” whereas the lower directions including the meaning of “diagonally low” are simply referred to as “low (lower)”.

First Embodiment

(10) A shaft sealing structure **12** according to a first embodiment of the present invention will be described with reference to FIGS. 1 to 5. In FIGS. 1 and 2, an angle seat valve **10** is shown in which the shaft sealing structure **12** is adopted. The angle seat valve **10** is used, for example, in order to control flowing of a fluid such as a chemical solution for cleaning a beverage container, or steam for carrying out steam sterilization on medical equipment or the like (hereinafter, simply referred to as a “fluid”). The angle seat valve **10** is particularly suitable for controlling a large flow rate. The angle seat valve **10** includes a valve mechanism unit and a drive mechanism unit. The valve mechanism unit is equipped with a valve body **38** and a valve rod (rod) **32**. The drive mechanism unit includes a cylinder tube **56** and a piston **58**.

(11) The valve body **38** includes a straight tubular shaped main pipe portion **40** having a fluid passage **40a** in the interior thereof, and an inclined pipe portion **42** that diagonally intersects the main pipe portion **40**. The inclined pipe portion **42** is formed integrally with the main pipe portion **40**. An axis CL2 of the inclined pipe portion **42** intersects an axis CL1 of the main pipe portion **40** at a predetermined angle (for example, 45 degrees). One end part of the main pipe portion **40** is equipped with an inlet port **40b**, and another end part of the main pipe portion **40** is equipped with an outlet port **40c**. The interior of the inclined pipe portion **42** is connected to the fluid passage **40a** of the main pipe portion **40**.

(12) A bottom wall of the main pipe portion **40** includes a raised portion **40d** thereon that protrudes toward the inclined pipe portion **42**. The valve body **38** includes an annular valve seat **40e**, which extends from an inner wall of the valve body **38** at a portion where the main pipe portion **40** and the inclined pipe portion **42** intersect each other at an obtuse angle, to the raised portion **40d**. The valve seat **40e** constitutes an annular sealing surface disposed perpendicularly to the axis CL2 of the inclined pipe portion **42**.

(13) A bottomed cylindrical holder **34** is attached to an upper part of the inclined pipe portion **42**. Specifically, a large-diameter hole portion **42b** having a female screw thread thereon is formed on an upper inner circumference of the inclined pipe portion **42** via a stepped portion **42a**. A large diameter part **34b** is formed on a lower outer circumference of the holder **34** via a first stepped portion **34a** (see FIG. 3). The holder **34** is fitted into the large diameter hole portion **42b** of the inclined pipe portion **42** until reaching a position where the large diameter part **34b** of the holder **34** abuts against the stepped portion **42a** of the inclined pipe portion **42**. Thereafter, a fixing cylindrical body **54** having a male screw thread thereon is inserted from an upper side of the inclined pipe portion **42** between the holder **34** and the inclined pipe portion **42**. Then, the fixing cylindrical body **54** is screwed-engaged with the inclined pipe portion **42**, and a distal end of the fixing cylindrical body **54** abuts against the first stepped portion **34a** of the holder **34**, whereby the holder **34** is fixed to the inclined pipe portion **42**. A sealing material **36a**, which is placed in contact with a wall surface of the large diameter hole portion **42b** of the inclined pipe portion **42**, is mounted on the large diameter part **34b** of the holder **34**.

(14) An upper part of the holder **34** extends upwardly from the inclined pipe portion **42**, and a later-described rod cover **66** is attached to an end part thereof. A first guide bush **24** and a second guide bush **26** are attached to an inner circumference of the holder **34**. The first guide bush **24** and the second guide bush **26** guide the valve rod **32** so as to be capable of moving in an axial direction. A pressing plate **28** is attached to a lower end of the holder **34**. A scraper **30** and a plurality of V-packings **14a** to **14c** are mounted between the pressing plate **28** and the first guide bush **24**. An auxiliary packing **22** is mounted between the first guide bush **24** and the second guide bush **26**. A description will be given later concerning the details of the shaft sealing structure **12**.

(15) A valve element **44** which is capable of abutting against the valve seat **40e** is connected to a lower end part of the valve rod **32**. Specifically, the valve element **44** includes an upwardly opening bottomed hole portion **44a** and a pair of pin holes **44b**. The hole portion **44a** extends along the axis CL2 of the valve element **44** (in common with the axis of the inclined pipe portion **42**). Each of the pin holes **44b** extends in a direction perpendicular to the axis CL2 of the valve element **44**, and opens on a wall surface of the hole portion **44a** and a side surface of the valve element **44**. On the other hand, the lower end part of the valve rod **32** includes a pin hole **32a**. The pin hole **32a** extends in a direction perpendicular to the axis CL2 of the valve rod **32**, and penetrates in a lateral direction through the valve rod **32**. By the lower end part of the valve rod **32** being inserted into the hole portion **44a** of the valve element **44**, and a connecting pin **48** being press-fitted into the pair of pin holes **44b** of the valve element **44** and the pin hole **32a** of the valve rod **32**, the valve element **44** is fixed to the valve rod **32**.

(16) The valve element **44** is arranged inside the inclined pipe portion **42**. When the valve element **44** is moved to a lowermost end together with the valve rod **32**, a seal member **46** which is mounted on the valve element **44** is placed in contact with the valve seat **40e**, and communication between the inlet port **40b** and the outlet port **40c** is interrupted. The direction in which the valve element **44** moves is inclined at a predetermined angle with respect to the direction in which the fluid flows from the inlet port **40b** toward the outlet port **40c**. Moreover, it should be noted that the members indicated by the reference numerals **50** and **52** are a nut and a washer for fixing the seal member **46** to the valve element **44**.

(17) The valve rod **32** is inserted into the first guide bush **24** and the second guide bush **26**, and extends upwardly from the holder **34**. An upper end part of the valve rod **32** is reduced in diameter, and is formed with a small diameter part **32b** having a male screw thread formed on a distal end thereof. A reinforcing plate **74**, the piston **58**, and a washer **76** are inserted over the small diameter part **32b** of the valve rod **32**, and a connecting rod **78** is screw-engaged with the small diameter part **32b** from an outer side thereof, whereby the piston **58** is attached to the upper end part of the valve rod **32**.

(18) The piston **58** is arranged inside the cylindrical cylinder tube **56**. The upper end side of the cylinder tube **56** is closed by a head cover **62** and a cap **64**. The lower end side of the cylinder tube **56** is closed by the rod cover **66**. A pressure chamber **70** into which air pressure can be introduced exists between the piston **58** and the rod cover **66**. Supplying and discharging of air to and from the pressure chamber **70** is carried out via a supply and discharge port **66a** of the rod cover **66**. Moreover, the reference numeral **67a** indicates a port for discharging air or the like within a chamber **67** that exists between the piston **58** and the head cover **62**.

(19) A pair of springs **72a** and **72b** that bias the piston **58** downward are arranged between the piston **58** and the head cover **62**. A piston packing **60** in sliding contact with the cylinder tube **56** is mounted on an outer circumference of the piston **58**. Further, a sealing material **36b**, which abuts against the rod cover **66**, and a sealing material **36c**, which is in sliding contact with the valve rod **32**, are mounted on the holder **34**. Due to such sealing materials, the pressure chamber **70** is maintained in an airtight state from the exterior, and the pressure chamber **70** is maintained in an airtight state from an internal space of the holder **34**.

(20) When air is introduced into the pressure chamber **70**, the piston **58** is driven upward in

opposition to a biasing force of the springs **72a** and **72b**, and the valve rod **32** and the valve element **44** are moved upward. On the other hand, when the air in the pressure chamber **70** is discharged, the piston **58** is driven downward by the biasing force of the springs **72a** and **72b**, and the valve rod **32** and the valve element **44** are moved downward. By adjusting the pressure of the air introduced into the pressure chamber **70**, the position of the piston **58** can be changed, and in accordance therewith, the degree at which the valve element **44** is opened can be adjusted.

(21) (Details of the Shaft Sealing Structure **12**)

(22) The V-packings **14a** to **14c** are made of a fluorine resin, and are superior in chemical resistance and steam resistance. As shown in FIG. **4**, each of the V-packings **14a** to **14c**, which are V-shaped in cross section, includes a thick-walled top portion **16**, a thin-walled inner side skirt portion **18** extending diagonally inward from the top portion **16**, and a thin-walled outer side skirt portion **20** extending diagonally outward from the top portion **16**. The top portion **16** includes an annular convex portion **16a** that protrudes to a side opposite to the inner side skirt portion **18** and the outer side skirt portion **20**. The plurality of V-packings **14a** to **14c** are arranged in such a manner that the respective top portions **16** thereof are positioned away from the valve element **44**, and the V-packings **14a** to **14c** are stacked in a direction of the axis CL2 of the valve rod **32** (in common with the axis of the inclined pipe portion **42**).

(23) The top portions **16** of adjacent ones of the V-packings **14a** to **14c** abut against each other by the convex portion **16a** of one of the adjacent top portions **16** abutting against a rear surface **16b** of the other of the adjacent top portions **16**. On the other hand, the inner side skirt portions **18** and the outer side skirt portions **20** of the adjacent ones of the V-packings **14a** to **14c** face each other with gaps therebetween, and do not interfere with each other. The inner side skirt portions **18** of the V-packings **14a** to **14c** slidably contact an outer surface of the valve rod **32** at distal end inner circumferential portions **18a** thereof. Further, the outer side skirt portions **20** of the V-packings **14a** to **14c** abut against an inner surface of the holder **34** at distal end outer circumferential portions **20a** thereof.

(24) According to the present embodiment, although a total of three V-packings **14a** to **14c** are provided including a lower V-packing **14a**, a middle V-packing **14b**, and an upper V-packing **14c**, the number of the V-packings is not necessarily limited to this feature.

(25) The auxiliary packing **22** is made of a rubber material such as FKM, and is superior in sealing performance. The auxiliary packing **22**, which is U-shaped in cross-section, includes an annular plate portion **22a**, an inner side cylindrical portion **22b** that extends from an inner circumferential portion of the annular plate portion **22a**, and an outer side cylindrical portion **22c** that extends from an outer circumferential portion of the annular plate portion **22a**. The inner side cylindrical portion **22b** of the auxiliary packing **22** slidably contacts the outer surface of the valve rod **32** on the inner circumference thereof. The outer side cylindrical portion **22c** of the auxiliary packing **22** abuts against the inner surface of the holder **34** on the outer circumference thereof.

(26) The first guide bush **24**, which is made of a resin such as PPS, is caught and positioned by a second stepped portion **34c** provided on the inner circumference of the holder **34**. The first guide bush **24** is equipped with a recessed portion **24a** in which the upper V-packing **14c** is accommodated, and abuts against the convex portion **16a** of the upper V-packing **14c** on the bottom surface of the recessed portion **24a**. Further, the first guide bush **24** is equipped with a protruding portion **24b** which enters between the inner side cylindrical portion **22b** and the outer side cylindrical portion **22c** of the auxiliary packing **22**.

(27) The second guide bush **26**, which is made of a resin such as PPS, is caught and positioned by a third stepped portion **34d** provided on the inner circumference of the holder **34**. A lower end surface of the second guide bush **26** abuts against the annular plate portion **22a** of the auxiliary packing **22**. The scraper **30** for removing foreign material, is made of a resin such as PEEK, and is positioned by the pressing plate **28**. A top portion of the scraper **30** abuts against the rear surface **16b** of the top portion **16** of the lower V-packing **14a**. The plurality of V-packings **14a** to **14c** are

sandwiched between the scraper **30** and the first guide bush **24**, and the adjacent ones of the top portions **16** are pressed against each other.

(28) The angle seat valve **10**, which includes the shaft sealing structure **12** in which the plurality of V-packings **14a** to **14c** and the auxiliary packing **22** are included, is configured in the manner described above. Next, a description will be given concerning actions and operations of the angle seat valve **10**.

(29) As shown in FIG. 2, a state in which the seal member **46** of the valve element **44** is placed in contact with the valve seat **40e** is defined as an initial state. In the initial state, air is not introduced into the pressure chamber **70**, and the piston **58** is positioned at the lowermost end due to the biasing force of the springs **72a** and **72b**. In this instance, when fluid is supplied to the inlet port **40b** of the valve body **38**, the fluid pressure acts on a lower end surface of the valve element **44**. However, since the biasing force of the springs **72a** and **72b** is greater than the force of the fluid pressure that presses the valve element **44** upward, the valve element **44** does not move.

(30) In this state, a non-illustrated switching valve is operated, and air is supplied from the supply and discharge port **66a** to the pressure chamber **70**. Upon doing so, a force that presses the piston **58** upward by the air pressure acting on the piston **58** exceeds the biasing force of the springs **72a** and **72b**, and the piston **58** moves upward. Then, the piston **58** moves to a position where the force that presses the piston **58** upward by the air pressure accumulated in the pressure chamber **70** and the force that presses the piston **58** downward by the reaction force of the contracted springs **72a** and **72b** are in balance (refer to FIG. 5). Moreover, strictly speaking, as the force that presses the piston **58** upward, there is included a force obtained by multiplying the pressure of the fluid acting on the valve element **44** by the cross-sectional area of the valve rod **32**.

(31) In this manner, the valve element **44** separates away from the valve seat **40e**, and the fluid supplied from the inlet port **40b** is discharged from the outlet port **40c** at a flow rate corresponding to the distance between the valve element **44** and the valve seat **40e** separated from each other. Thereafter, the switching valve is operated, and the air in the pressure chamber **70** is discharged to the exterior from the supply and discharge port **66a**. Upon doing so, the piston **58** moves to the lowermost end due to the biasing force of the springs **72a** and **72b**, and the seal member **46** of the valve element **44** is placed in contact with the valve seat **40e**. Consequently, communication between the inlet port **40b** and the outlet port **40c** is interrupted, and discharging of the fluid from the outlet port **40c** is stopped.

(32) The fluid supplied from the inlet port **40b** fills the inclined pipe portion **42** each time that the valve element **44** separates away from the valve seat **40e**. By the shaft sealing structure **12** in which the plurality of V-packings **14a** to **14c** and the auxiliary packing **22** are included, the fluid inside the inclined pipe portion **42** is prevented from leaking out to the exterior.

(33) More specifically, the lower V-packing **14a** receives the pressure of the fluid inside the inclined pipe portion **42**, the thin-walled inner side skirt portion **18** is elastically deformed, and the pressure at which the distal end inner circumferential portion **18a** abuts against the outer surface of the valve rod **32** increases. Further, the thin-walled outer side skirt portion **20** is elastically deformed, and the pressure at which the distal end outer circumferential portion **20a** abuts against the inner surface of the holder **34** increases. Consequently, the lower V-packing **14a** exhibits a high sealing performance, and the fluid in the inclined pipe portion **42** is effectively suppressed from passing through the lower V-packing **14a**. In the case that a slight amount of fluid passes through the lower V-packing **14a**, the auxiliary packing **22** prevents the fluid from leaking to the exterior.

(34) When the angle seat valve **10** has been used over a certain period of time, wearing of the lower V-packing **14a** progresses, whereby the lower V-packing **14a** becomes incapable of exhibiting a sufficient sealing performance, and its role and functionality come to an end. When this happens, instead of the lower V-packing **14a**, the middle V-packing **14b**, which receives the pressure of the fluid inside the inclined pipe portion **42**, becomes capable of exhibiting such a sealing performance. Furthermore, when the angle seat valve **10** has been used over a certain period of time, wearing of

the middle V-packing **14b** progresses. Then, instead of the middle V-packing **14b**, the upper V-packing **14c** becomes capable of exhibiting such a sealing performance. In this manner, the sealing action is sequentially handed over from the downwardly positioned V-packing to the upwardly positioned V-packing. It should be noted that wearing of the V-packings **14a** to **14c** progresses particularly at the distal end inner circumferential portions **18a** of the inner side skirt portions **18** that repeatedly slide in contact with the reciprocating valve rod **32**.

(35) Since the inner side skirt portions **18** and the outer side skirt portions **20** of the V-packings **14a** to **14c** are formed to be thin-walled, they are easily elastically deformed when subjected to the pressure of the fluid inside the inclined pipe portion **42**, and are capable of exhibiting satisfactory sealing performance. Further, since the adjacent ones of the V-packings **14a** to **14c** abut against each other at the top portions **16** which are formed to be thick-walled and have highly rigidity, the gaps between the inner side skirt portions **18** and the outer side skirt portions **20** of the adjacent ones of the V-packings **14a** to **14c** are reliably retained. In addition, since the inner side skirt portions **18** and the outer side skirt portions **20** of the adjacent ones of the V-packings **14a** to **14c** do not interfere with each other, wearing of the plurality of V-packings **14a** to **14c** does not progress simultaneously, and satisfactory sealing performance can be therefore maintained over a prolonged period of time.

(36) The shaft sealing structure **12** according to the present embodiment includes the plurality of V-packings **14a** to **14c** that are formed of the thick-walled top portions **16**, the thin-walled inner side skirt portions **18**, and the thin-walled outer side skirt portions **20**. In addition, since the adjacent ones of the inner side skirt portions **18** and the outer side skirt portions **20** do not interfere with each other, satisfactory sealing performance can be maintained over a prolonged period of time.

(37) According to the present embodiment, although the V-packings **14a** to **14c** are made of a fluorine resin, the V-packings may be made of a resin other than a fluorine resin, rubber, an elastomer or the like, as long as the material has a predetermined elasticity. Further, in the case that the V-packings **14a** to **14c** are made of a fluorine resin, the auxiliary packing **22** which is made of a rubber material is useful for enhancing the sealing performance. For example, in the case that the V-packings **14a** to **14c** are made of a rubber material, the auxiliary packing **22** need not necessarily be provided. Although in describing the shaft sealing structure **12** according to the present embodiment, an example in which the shaft sealing structure **12** is applied to an angle seat valve for fluids such as a chemical solution and steam, the shaft sealing structure **12** can be applied to various devices (valves) for fluids such as compressed air.

Second Embodiment

(38) A shaft sealing structure **80** according to a second embodiment of the present invention will be described with reference to FIGS. **6** and **7**. The shaft sealing structure **80** will be described as being applied to the same angle seat valve **10** as in the first embodiment. Further, the same reference numerals designate the same or equivalent constituent elements as those of the shaft sealing structure **12** according to the first embodiment, and detailed description thereof is omitted.

(39) The shaft sealing structure **80** includes a plurality of U-packings **82a** to **82c** made of a fluorine resin. As shown in FIG. **7**, each of the U-packings **82a** to **82c**, which are U-shaped in cross section, includes a thick-walled top portion **84**, a thin-walled inner side skirt portion **86** extending inwardly and in an axial direction from the top portion **84**, and a thin-walled outer side skirt portion **88** extending outwardly and in the axial direction from the top portion **84**. Moreover, the “axial direction” referred to herein is a direction parallel to a central axis of the U-packings **82a** to **82c** which coincides with the axis CL2 of the valve rod **32**.

(40) The plurality of U-packings **82a** to **82c** are arranged in such a manner that the respective top portions **84** thereof are positioned away from the valve element **44**, and the U-packings **82a** to **82c** are stacked in a direction of the axis CL2 of the valve rod **32**. According to the present embodiment, as shown in FIG. **6**, although a total of three U-packings **82a** to **82c** are provided including a lower U-packing **82a**, a middle U-packing **82b**, and an upper U-packing **82c**, the

number of the U-packings is not necessarily limited to this feature.

(41) The inner side skirt portion **86** of each of the U-packings **82a** to **82c** is provided with a plurality of inwardly protruding annular ribs **86a**. The outer side skirt portion **88** of each of the U-packings **82a** to **82c** is provided with a plurality of outwardly protruding annular ribs **88a**. The inner side skirt portions **86** and the outer side skirt portions **88**, which are formed to be thin-walled, are elastically deformed when subjected to the pressure of the fluid inside the inclined pipe portion **42**. Then, the inner side skirt portions **86** slidably contact the outer surface of the valve rod **32** at the distal ends of the plurality of ribs **86a**. Further, the outer side skirt portions **88** abut against the inner surface of the holder **34** at the distal ends of the plurality of ribs **88a**. Consequently, at the contact portions thereof, it is possible to secure a surface pressure and a contact area that are greater than or equal to a predetermined value, and satisfactory sealing performance can be exhibited.

(42) Further, end surfaces **84a** and rear surfaces **84b** of the top portions **84** of the U-packings **82a** to **82c** are both flat surfaces. The top portions **84** of adjacent ones of the U-packings **82a** to **82c** abut against each other in a state in which the end surface **84a** of one of the adjacent top portions **84** abuts against the rear surface **84b** of the other of the adjacent top portions **84**, in other words, the two flat surfaces abut against each other. Consequently, the central axes of the plurality of stacked U-packings **82a** to **82c** are stably kept coinciding with each other.

(43) Further, the length in the axial direction of the top portions **84** of the U-packings **82a** to **82c** is longer than a distance from the rear surfaces **84b** of the top portions **84** to the distal ends of the inner side skirt portions **86** or the distal ends of the outer side skirt portions **88**. Stated otherwise, the thickness **T** of the top portions **84** is greater than the depth **D** of the recesses formed by the inner side skirt portions **86** and the outer side skirt portions **88**. Consequently, the inner side skirt portions **86** and the outer side skirt portions **88** of the adjacent ones of the U-packings **82a** to **82c** do not interfere with each other.

(44) When the angle seat valve **10** has been used over a certain period of time, wearing of the lower U-packing **82a** progresses, whereby the lower U-packing **82a** becomes incapable of exhibiting a sufficient sealing performance, and its role and functionality come to an end. When this happens, the middle U-packing **82b**, which receives the pressure of the fluid inside the inclined pipe portion **42**, becomes capable of exhibiting such a sealing performance. Furthermore, when the angle seat valve **10** has been used over a certain period of time, wearing of the middle U-packing **82b** progresses. Then, the upper U-packing **82c** becomes capable of exhibiting such a sealing performance. In this manner, the sealing action is sequentially handed over from the downwardly positioned U-packings to the upwardly positioned U-packings, and wearing of the plurality of U-packings **82a** to **82c** does not progress simultaneously.

(45) The shaft sealing structure **80** according to the present embodiment includes the plurality of U-packings **82a** to **82c** that are formed of the thick-walled top portions **84**, the thin-walled inner side skirt portions **86**, and the thin-walled outer side skirt portions **88**. In addition, since the adjacent ones of the inner side skirt portions **86** and the outer side skirt portions **88** do not interfere with each other, satisfactory sealing performance can be maintained over a prolonged period of time.

(46) According to the present embodiment, although the U-packings **82a** to **82c** are made of a fluorine resin, the U-packings may be made of a resin other than a fluorine resin, rubber, an elastomer or the like, as long as the material has a predetermined elasticity.

(47) (Reference Example)

(48) A shaft sealing structure **90** in the form of a reference example is shown in FIG. **8**. This shaft sealing structure **90** differs from the shaft sealing structure **12** according to the first embodiment in that only one V-packing is provided therein. Moreover, the same reference numerals designate the same or equivalent constituent elements as those of the shaft sealing structure **12** according to the first embodiment. Hereinafter, a description will be given concerning the shaft sealing structure **90**.

(49) The first guide bush **24** and the second guide bush **26** are attached to the inner circumference of the holder **34**. The first guide bush **24** and the second guide bush **26** guide the valve rod **32** so as

to be capable of moving in an axial direction. The pressing plate **28** is attached to the lower end of the holder **34**. The scraper **30** and the single V-packing **14** are mounted between the pressing plate **28** and the first guide bush **24**. A rubber seal packing **22** is mounted between the first guide bush **24** and the second guide bush **26**.

(50) The V-packing **14** is made of a fluorine resin such as PTFE or PFA, and is superior in chemical resistance, heat resistance, and steam resistance. The V-packing **14**, which is V-shaped in cross section, includes the top portion **16**, the thin-walled inner side skirt portion **18** extending diagonally inward from the top portion **16**, and the thin-walled outer side skirt portion **20** extending diagonally outward from the top portion **16**. The top portion **16** includes the annular convex portion **16a** that protrudes to a side opposite to the inner side skirt portion **18** and the outer side skirt portion **20**. The inner side skirt portion **18** of the V-packing **14** slidably contacts the outer surface of the valve rod **32** at the distal end inner circumferential portion **18a** thereof. The outer side skirt portion **20** of the V-packing **14** abuts against the inner surface of the holder **34** at the distal end outer circumferential portion **20a** thereof.

(51) The rubber seal packing **22** is made of a rubber material such as NBR, HNBR, EPDM, FKM, or silicone, and is superior in sealing performance. The rubber seal packing **22**, which is U-shaped in cross-section, includes the annular plate portion **22a**, the inner side cylindrical portion **22b** that extends from the inner circumferential portion of the annular plate portion **22a**, and the outer side cylindrical portion **22c** that extends from the outer circumferential portion of the annular plate portion **22a**. The inner side cylindrical portion **22b** of the rubber seal packing **22** slidably contacts the outer surface of the valve rod **32** on the inner circumference thereof. The outer side cylindrical portion **22c** of the rubber seal packing **22** abuts against the inner surface of the holder **34** on the outer circumference thereof.

(52) The first guide bush **24** is made of a super engineering plastic material having a high mechanical strength, a high sliding ability, a heat resistance, and a low wear rate. As specific examples of such a super engineering plastic material, there may be cited PPS, PVDF, PA, PC, PE, POM, PEI, PPO, PET, PTFE, PEEK, PAI, PI, and PBI. The first guide bush **24** is caught by the second stepped portion **34c** provided on the inner circumference of the holder **34**. The first guide bush **24** is equipped with the recessed portion **24a** in which the V-packing **14** is accommodated, and abuts against the convex portion **16a** of the V-packing **14** on the bottom surface of the recessed portion **24a**. Further, the first guide bush **24** is equipped with the protruding portion **24b** which enters between the inner side cylindrical portion **22b** and the outer side cylindrical portion **22c** of the rubber seal packing **22**.

(53) The first guide bush **24** includes a plurality of first grease reservoirs **24c** that open on the bottom surface of the recessed portion **24a**, and an annular second grease reservoir **24d** that opens toward the rubber seal packing **22**. The plurality of first grease reservoirs **24c** are uniformly arranged in the circumferential direction. The grease that is stored in the plurality of first grease reservoirs **24c** is supplied to the V-packing **14**. The grease that is stored in the second grease reservoir **24d** is supplied to the rubber seal packing **22**. Since the first guide bush **24** is disposed above the V-packing **14**, the first guide bush **24** is protected from the fluid by the V-packing **14**, and the durability thereof is enhanced.

(54) The second guide bush **26** is made of a super engineering plastic material having a high mechanical strength, a high sliding ability, a heat resistance, and a low wear rate. The second guide bush **26** is caught by the third stepped portion **34d** provided on the inner circumference of the holder **34**. The lower end surface of the second guide bush **26** abuts against the annular plate portion **22a** of the rubber seal packing **22**. The second guide bush **26** includes a grease reservoir **26a** on an inner circumferential surface thereof that faces the valve rod **32**. Since the valve rod **32** is supported over a sufficient length by the first guide bush **24** and the second guide bush **26**, the stability with which the valve rod **32** is supported is improved.

(55) The scraper **30** is made of a super engineering plastic material having a heat resistance, a

steam resistance, a chemical resistance, a high mechanical strength, a high sliding ability, and a low wear rate. The scraper **30** prevents foreign material such as sludge within the fluid that adheres to the valve rod **32** from entering the interior of the shaft sealing structure **90**. The scraper **30** is positioned by the pressing plate **28**. An upper end part of the scraper **30** abuts against the rear surface **16b** of the top portion **16** of the V-packing **14**.

(56) The V-packing **14** is arranged between the first guide bush **24** and the scraper **30**. The top portion **16** of the V-packing **14** is sandwiched between the bottom surface of the recessed portion **24a** of the first guide bush **24** and the upper end part of the scraper **30**. The inner side skirt portion **18** and the outer side skirt portion **20** of the V-packing **14** can be freely deformed inside a space formed between the first guide bush **24** and the scraper **30**. Since the inner side skirt portion **18** and the outer side skirt portion **20** of the V-packing **14** are formed to be thin-walled, they are easily elastically deformed when subjected to the pressure of the fluid, and exhibit satisfactory sealing performance.

(57) Since the shaft sealing structure **90** includes only one V-packing **14**, the durability of the seal is somewhat inferior to that of the aforementioned shaft sealing structure **12** in which the plurality of V-packings **14a** to **14c** are included. However, the configuration in which the first guide bush **24** and the scraper **30** are included is useful in a similar manner as the aforementioned shaft sealing structure **12**, and such a configuration is practical in consideration of cost effectiveness.

(58) The present invention is not limited to the above-described embodiments, and various configurations can be adopted therein without departing from the essence and gist of the present invention.

Claims

1. A shaft sealing structure arranged between a rod configured to be driven within a predetermined fluid and move in an axial direction, and a holder in which the rod is accommodated, the shaft sealing structure comprising: a V-packing; a first guide bush attached to an inner circumference of the holder; and a scraper arranged at an end of the holder, wherein the V-packing is formed of a top portion formed to be thick-walled, an inner side skirt portion formed to be thin-walled and extending diagonally inward from the top portion, and an outer side skirt portion formed to be thin-walled and extending diagonally outward from the top portion, the inner side skirt portion slidably contacts an outer surface of the rod at a distal end of the inner side skirt portion, and the outer side skirt portion abuts against an inner surface of the holder at a distal end of the outer side skirt portion, and the top portion is sandwiched between the scraper and the first guide bush, and the inner side skirt portion and the outer side skirt portion are freely deformable inside a space formed between the first guide bush and the scraper.

2. The shaft sealing structure according to claim 1, wherein the top portion includes a convex portion configured to protrude to a side opposite to the inner side skirt portion and the outer side skirt portion, the first guide bush is equipped with a recessed portion in which the upper V-packing is accommodated, and abuts against the convex portion of the V-packing on a bottom surface of the recessed portion, and an end of the scraper abuts against a rear surface of the top portion of the V-packing.

3. The shaft sealing structure according to claim 2, wherein the first guide bush includes a first grease reservoir that opens on the bottom surface of the recessed portion.

4. The shaft sealing structure according to claim 1, further comprising a second guide bush attached to the inner circumference of the holder, wherein a rubber seal packing is mounted between the first guide bush and the second guide bush.

5. The shaft sealing structure according to claim 4, wherein the rubber seal packing includes an annular plate portion, an inner side cylindrical portion that extends from an inner circumferential portion of the annular plate portion, and an outer side cylindrical portion that extends from an outer

circumferential portion of the annular plate portion, and an inner circumference of the inner side cylindrical portion slidably contacts the outer surface of the rod, and an outer circumference of the outer side cylindrical portion slidably contacts the inner surface of the holder.

6. The shaft sealing structure according to claim 4, wherein the first guide bush includes a grease reservoir that opens toward the rubber seal packing.

7. The shaft sealing structure according to claim 4, wherein the second guide bush includes a grease reservoir on an inner circumferential surface thereof that faces the rod.

8. The shaft sealing structure according to claim 1, wherein the V-packing is made of a fluorine resin.

9. A valve adopting the shaft sealing structure according to claim 1.
